



Summary

Alzheimer's disease (AD) progresses along a continuum from cognitively normal (CN) aging through mild cognitive impairment (MCI) to dementia, and this progression is widely believed to reflect a gradual failure of the hippocampal memory circuit, expressed both in how strongly that circuit communicates with the rest of the brain and in how much of its tissue remains. We examined both expressions of this failure in the same population, using two pipelines built independently for this purpose and applied here, for the first time, to the full available ADNI cohort rather than to development samples.

Taken together, the two independently developed analyses converge on a single picture - connectivity and morphometry of the same hippocampal circuit degrade together across the AD spectrum, and this degradation is detectable at the individual level from resting-state connectivity alone. This convergence, obtained from two separate pipelines and two only partially overlapping subsamples of the same cohort, is the central result we present.

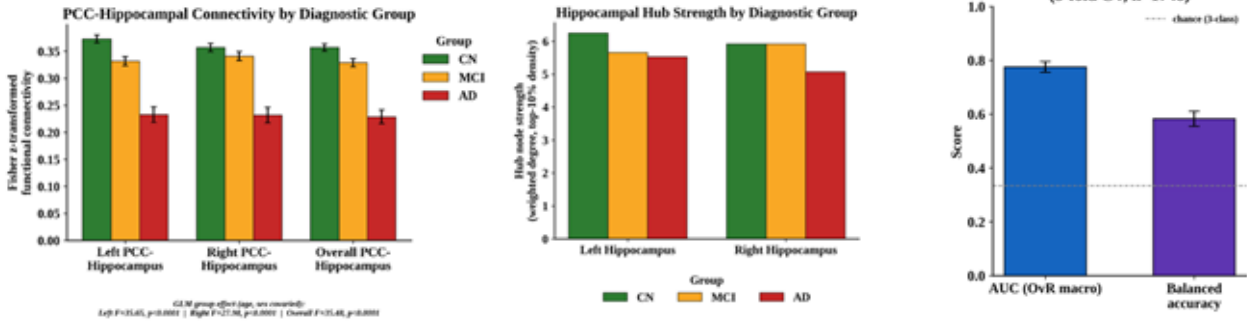
session 1: resting-state functional connectivity

Sample: all ADNI resting-state scans with both a preprocessed BOLD volume and confounds file, restricted to CN/MCI/AD — 1,749 scans from 945 participants (824 CN/510, 686 MCI/353, 239 AD/150).

Signals were extracted with a combined 202-region atlas: Schaefer-200 cortical parcellation (Yeo 7 networks) [14,15] plus left/right hippocampus from the Harvard-Oxford atlas [16]. Motion scrubbing removed volumes with FD > 0.5 mm or RMSD > 0.3 mm (scans with <20 clean volumes excluded); regional time series were band-pass filtered (0.01–0.09 Hz) and residualized against motion and white-matter/CSF signals.

From each subject's Fisher z-transformed 202×202 correlation matrix we derived: (1) PCC-hippocampal connectivity, by hemisphere and averaged, plus a laterality index; (2) hippocampal/PCC hub strength (weighted degree at top-10%-density threshold); (3) a 3-class L2 logistic regression classifier on all pairwise edges, evaluated by 5-fold stratified CV (macro AUC, balanced accuracy), with top-contributing edges extracted as candidate biomarkers. Group effects were tested with an age/sex-adjusted GLM. One scan had 101/202 zero-variance ROI signals, producing an undefined correlation matrix — it was excluded only at the classifier stage.

Results



PCC-hippocampal connectivity declined monotonically from CN to AD in both hemispheres (left: 0.373→0.332→0.233; right: 0.357→0.341→0.232; bilateral average: 0.357→0.329→0.229; all GLM $p < 0.0001$). Laterality showed no group effect ($p = 0.30$) — both hemispheres decline together (Figure 1). Hippocampal hub strength followed the same pattern (left 6.25→5.66→5.53; right 5.92→5.92→5.07), with the largest drop on the right between MCI and AD. PCC hub strength changed little by comparison (Figure 2), suggesting the loss of centrality is hippocampus-specific.

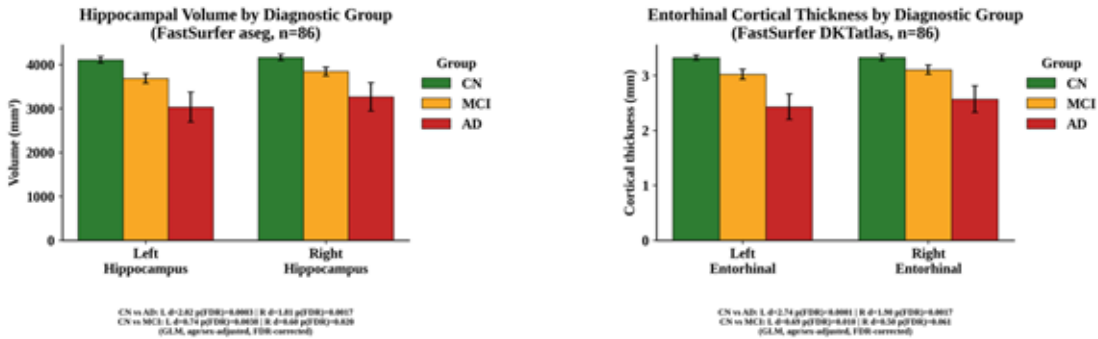
The classifier, trained on 1,748 scans after excluding the corrupted scan, reached AUC = 0.775 (SD 0.020) and balanced accuracy = 0.583 (SD 0.028) — well above the 0.333 chance level (Figure 3). The top edge was left-right limbic temporal pole connectivity (importance 0.163); the right hippocampus appeared directly at rank 8 (Table 2).

Functional disconnection between the DMN hub and hippocampus, and loss of hippocampal network centrality, both scale with clinical severity and are reproducible enough to support individual-level classification from connectivity alone.

session 2: validating medial temporal lobe hub degeneration in AD progression using T1 MRI

Sample: 86 ADNI participants with FastSurfer-processed T1 scans and complete phenotypic data: 41 CN (age 72.0 ± 4.1), 39 MCI (72.3 ± 7.6), 6 AD (74.8 ± 9.1). All passed QC. FastSurfer [5,7] segmented each scan and produced a DKT cortical parcellation [6]. We used hippocampal volume (from subcortical segmentation) and entorhinal/parahippocampal volume, thickness, and surface area (from the cortical parcellation), bilaterally — the earliest cortical sites of AD-related tau pathology [2,3]. For each region we tested CN-vs-MCI, MCI-vs-AD, and CN-vs-AD with Welch's t-test/Cohen's d and an age/sex-adjusted GLM (region ~ group + age + sex), FDR-corrected across regions.

Results



Hippocampal volume declined CN→MCI→AD in both hemispheres (left: 4108→3683→3034 mm³, 26% loss; right: 4162→3841→3265 mm³, 22% loss). CN-vs-AD and CN-vs-MCI contrasts were significant after correction (left $d=2.02/0.74$, $p_{FDR}=0.0003/0.0038$; right $d=1.81/0.60$, $p_{FDR}=0.0017/0.020$). MCI-vs-AD was directionally consistent ($d=0.95/0.90$) but underpowered ($p_{FDR} \approx 0.10$) given $n=6$ in the AD group (Figure 4).

Entorhinal thickness showed the largest effect in the study: left CN→MCI→AD 3.32→3.02→2.43 mm, CN-vs-AD $d=2.74$ ($p_{FDR}<0.0001$); right $d=1.90$ ($p_{FDR}=0.0017$). CN-vs-MCI was significant on the left ($d=0.69$, $p_{FDR}=0.010$) (Figure 5).

Table 3. Structural Group Comparisons

Region	Contrast	Cohen's d	p (FDR)
Left hippocampal volume	CN vs AD	2.02	0.0003
Right hippocampal volume	CN vs AD	1.81	0.0017
Left hippocampal volume	CN vs MCI	0.74	0.0038
Right hippocampal volume	CN vs MCI	0.60	0.020
Left entorhinal thickness	CN vs AD	2.74	<0.0001
Right entorhinal thickness	CN vs AD	1.90	0.0017
Left entorhinal thickness	CN vs MCI	0.69	0.010
Left entorhinal gray volume	CN vs AD	1.77	0.0012
Left parahippocampal thickness	CN vs AD	2.22	0.0003

Entorhinal and parahippocampal gray volume followed the same gradient (full results in Table 3).

Effects are consistent across every region despite the small AD group, indicating the CN-vs-AD contrast is robust and that the null MCI-vs-AD result reflects power, not a true plateau. The structural gradient really does existed in our experiment, and it mirrors the functional decline reported in session 1 (rs-fMRI; see the first page of the poster) over the same hippocampal circuit.

Reflection of our work & BCSC Term 2

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